

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

mediShop

Online Pharmacy & Digital Healthcare Platform

(Bangladesh Market • MedEasy-inspired with UX Improvements)

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Primary Focus	Frontend Phase-1 (Client Demo)
Tech Stack	Next.js 15 • TypeScript • Tailwind • shadcn/ui
State Management	TanStack Query + Redux Toolkit
Theme Color	Blue (#1D4ED8)
Typography	Inter (body) + Roboto Slab (titles)

Prepared for Client Presentation & Development Team
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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the complete functional, non-functional, UI/UX, and technical requirements for mediShop — a modern online pharmacy and digital healthcare platform targeted at the Bangladesh market. The platform is inspired by MedEasy.health but incorporates significant UX, performance, trust, and localization improvements.

The primary goal of Phase-1 is to deliver a polished, interactive frontend prototype that can be demonstrated to the client. All core user flows (browse, search, cart, checkout UI, authentication, prescription upload) must be fully functional with mock data.

1.2 Scope

In Scope (Phase-1 Frontend): Responsive web application, Homepage with Hero Slider & Category Sidebar, Global Search, Product Listing with advanced Filters & Sorting, Product Cards, Cart Drawer + Cart Page, Checkout Page with Payment method selection (UI only), Authentication Modal, Prescription Upload CTA & flow UI, Trust badges, Floating Call/WhatsApp buttons, Bangla-first content support.

Out of Scope (Phase-1): Real backend APIs, actual payment gateway integration, order management admin panel, real-time order tracking, video consultation, push notifications, native mobile apps.

1.3 Definitions & Acronyms

Term	Definition
SRS	Software Requirements Specification
UI / UX	User Interface / User Experience
CTA	Call To Action
OTC	Over-The-Counter (medicines that do not require prescription)
DGDA	Directorate General of Drug Administration (Bangladesh)
FR	Functional Requirement
NFR	Non-Functional Requirement
mediShop	Project name of this online pharmacy platform

1.4 References

- MedEasy.health – Reference platform for layout & feature inspiration
- Previous UX improvement recommendations (Bangladesh context)
- Client-provided technology constraints (Next.js, TypeScript, Tailwind, shadcn/ui, TanStack Query, Redux)

2. Overall Description

2.1 Product Perspective

mediShop is a consumer-facing digital healthcare platform that enables users in Bangladesh to order authentic medicines (prescription and OTC), upload prescriptions for pharmacist verification, browse health & wellness products, and access entry points for doctor consultations. It prioritizes mobile users, low-bandwidth environments, Bangla language, and strong trust signals.

2.2 Product Functions (High-Level)

- Browse medicines and healthcare products by category with visual sidebar navigation
- Powerful global search supporting Bangla, English and common misspellings
- Upload prescription and track verification status (UI)
- Add products to cart, manage quantities, view free-delivery progress
- Complete checkout flow including address, delivery options and payment method selection
- User authentication via modal (Login / Register / OTP-ready)
- View deals, discounts, trust badges and how-it-works sections
- Floating Call & WhatsApp order buttons for users who prefer traditional ordering

2.3 User Classes & Characteristics

User Class	Characteristics
Guest User	Can browse, search, view products, open cart drawer. Must login/register to checkout or save addresses.
Registered Customer	Full access to cart, checkout, order history (future), saved addresses, prescription upload.
Chronic Patient	Priority experience – easy re-order, monthly packs, auto-refill entry points (future).
Elderly / Low-literacy	Large touch targets, icon-heavy UI, clear Bangla labels, Call/WhatsApp fallback.

2.4 Operating Environment

- Modern browsers: Chrome, Firefox, Safari, Edge (latest 2 versions)
- Mobile devices: Android 8+ and iOS 13+ (primary target – mid-range Android phones)
- Network: Optimized for 3G/4G mobile data (low bandwidth)

2.5 Design & Implementation Constraints

Mandatory Technology Stack (Frontend Phase-1):

Layer	Technology
Framework	Next.js 15 (App Router)
Language	TypeScript (strict mode)
Styling	Tailwind CSS
UI Components	shadcn/ui
Server / Async State	TanStack Query (React Query)
Client State	Redux Toolkit
Forms	React Hook Form + Zod
Icons	Lucide React

Layer	Technology
Animations	Framer Motion (recommended)
Notifications	Sonner
Body Font	Inter
Title / Header Font	Roboto Slab (Serif)
Theme Color	Blue (#1D4ED8)

2.6 Assumptions & Dependencies

- Backend APIs will be provided in later phases; Phase-1 uses realistic mock data.
- Bangla font (Noto Sans Bengali or equivalent) will be properly loaded and subsetting.
- Product images will be optimized (Next.js Image + WebP/AVIF).
- Client will provide final brand logo assets and exact copy if different from defaults.
- Theme color Blue (#1D4ED8) and typography (Inter + Serif) are confirmed for the design system.

3. System Features & Functional Requirements

3.1 Global Navigation Bar

FR-01 — Navbar Structure

A sticky top navigation bar shall be present on all pages with the following layout:

- Left: mediShop Logo (clickable → Homepage)
- Center: Global Search bar (full-width on desktop; expandable on mobile)
- Right: Cart icon with live badge (item count) + Account dropdown (logged-in) or Login button (guest)
- Login button opens Authentication Modal (does not navigate to a separate page).
- On mobile a hamburger menu may collapse secondary links; Cart and Account remain visible.

3.2 Authentication (Modal)

FR-02 — Auth Modal

Authentication shall occur inside a modal dialog (shadcn Dialog). Supported flows: Login, Register, Forgot Password. Preferred method for Bangladesh: Phone number + OTP (UI ready). Email/Password also supported. Social login placeholders (Google) may be shown. On successful authentication the modal closes and the Navbar updates to show the Account dropdown.

3.3 Homepage

FR-03 — Homepage Layout

The homepage shall follow a MedEasy-inspired structure enhanced with the recommended UX improvements.

Desktop layout:

Sticky Navbar (Logo Search Cart + Account)		
Category Sidebar (Left)	Hero Slider	
	Trust Badges / Quick Stats	
	Featured Categories (Icon Grid)	
	Exclusive Deals / Popular Products	
	Prescription Upload CTA Banner	
	How it Works + Doctor Consultation CTA	
	Additional Product Sections	

FR-04 — Hero Slider

Auto-playing hero slider (4-6 slides) with manual controls. Each slide contains strong Bangla + English messaging and primary CTAs (Upload Prescription, Shop Now, Consult Doctor, etc.). Fully responsive.

FR-05 — Category Sidebar

Vertical sidebar listing main categories with icons. Clicking a category filters or navigates to the corresponding product listing. Active state is clearly highlighted. On mobile the sidebar becomes a horizontal scrollable list or opens as a bottom sheet / drawer.

FR-06 — Product Sections on Homepage

Multiple product carousels or grids: Exclusive Deals, Popular / Fast-Moving, Women's Choice, Diabetic Care, Baby Care, Sexual Wellness, Personal Care, etc. Each Product Card displays: image, discount badge, brand, name, current price + strikethrough MRP, and "Add to Cart" button. Skeleton loaders while data is fetched.

FR-07 — Floating Action Buttons

Persistent floating buttons (especially on mobile) for "Call to Order" and "WhatsApp Order" to support users who prefer traditional ordering channels.

3.4 Search

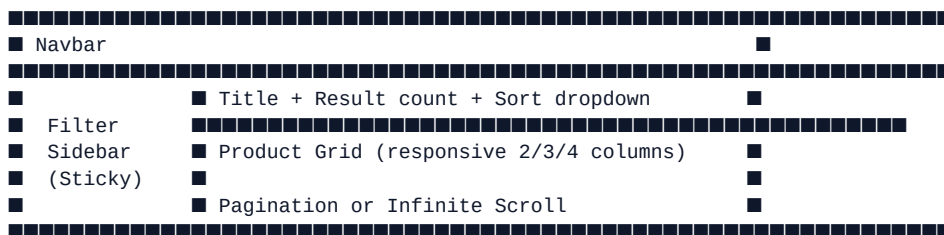
FR-08 — Global Search

Global search in the Navbar supports Bangla, English and common misspellings. Instant suggestions powered by TanStack Query. Shows recent searches and popular searches. Search results page includes the same Filter Sidebar and Sorting controls as product listing pages.

3.5 Product Listing Pages (Filters, Sorting, Search)

Applies to: /products, /category/[slug], /search?q=...

FR-09 — Listing Page Layout



FR-10 — Filter Sidebar (Left)

Mandatory filters:

- Categories (multi-select or hierarchical)
- Brand (searchable checkbox list)
- Price Range (slider + min/max inputs)
- Discount percentage (10%+, 20%+, 30%+ ...)
- Availability (In Stock)
- Prescription Required (Yes / No / OTC)
- "Clear All Filters" action + active filter chips displayed above the grid
- Mobile: Filter button opens a bottom sheet / drawer

FR-11 — Sorting

Sort dropdown options: Popularity (default), Price: Low to High, Price: High to Low, Discount: High to Low, Newest First, Name: A-Z.

FR-12 — Product Grid Behavior

Responsive columns (Mobile 2, Tablet 3, Desktop 4). Skeleton loading states. "Add to Cart" works directly from the card. Pagination or infinite scroll (configurable).

3.6 Product Detail Page

FR-13 — Product Detail (Basic for Phase-1)

Displays product images, name (Bangla + English preferred), brand, price & MRP, discount, stock status, delivery estimate, short description, and Add to Cart / Buy Now actions. Related products or "Same medicine different brands" section is recommended.

3.7 Cart System (Icon, Drawer, Full Page)

FR-14 — Cart Icon in Navbar

Cart icon placed immediately beside the Account icon/button. Displays a badge with the current item count. Badge updates in real time from Redux state.

FR-15 — Cart Sidebar (Drawer)

Clicking the Cart icon opens a right-side Drawer (shadow Sheet). Contents:

- List of cart items (image, name, price, quantity steppers, remove)
- Subtotal
- Free-delivery progress bar (e.g., "Add X more for free delivery")
- Primary actions: "View Cart" (navigates to /cart) and "Checkout"
- Beautiful empty-cart state

FR-16 — Dedicated Cart Page (/cart)

Full-page cart experience with quantity controls, remove, clear cart, optional coupon/promo code field (UI only in Phase-1), sticky order summary on desktop, and a prominent "Proceed to Checkout" button.

3.8 Checkout & Payment Flow

FR-17 — Checkout Page (/checkout)

The checkout page shall contain the following sections (single-page or multi-step):

1. Delivery Address — Saved addresses (logged-in users) + Add new address form (Name, Phone, Area, Full Address, District). Default address selection.
2. Delivery Options — Same-day (selected cities) / Standard (2–3 days). Delivery charge calculated and shown.
3. Order Summary — Line items, delivery charge, discounts, Grand Total.
4. Payment Method Selection — bKash, Nagad, Rocket, Card, Cash on Delivery. Clear icons and selection state.
5. Place Order — Primary CTA that triggers the payment simulation.

FR-18 — Payment Flow (Phase-1 Simulation)

Selecting a payment method reveals the corresponding UI (e.g., bKash wallet number field). "Pay Now" shows a loading state then a Success or Failed mock result. On success the user is redirected to an Order Confirmation page. No real payment gateway is integrated in Phase-1.

3.9 Prescription Upload

FR-19 — Prescription Upload Flow

Prominent CTA on Homepage and dedicated entry points. Step-by-step visual guidance: 1. Capture clear photo → 2. Upload → 3. Pharmacist verifies → 4. Confirmation call / status. Progress awareness UI. Multiple prescription upload and previous-prescription reuse are desirable.

3.10 Trust & Credibility Sections

FR-20 — Trust Signals

The following elements must be clearly visible (especially above the fold on Homepage):

- DGDA License number

- "100% Authentic Medicines" messaging
- "Supported by Startup Bangladesh" badge (if applicable)
- Delivery information (Same-day Dhaka, 2–3 days nationwide)
- Customer count / social proof / reviews
- SSL and secure-payment badges

4. Non-Functional Requirements

ID	Requirement	Priority
NFR-01	Mobile-first and fully responsive across all breakpoints	Must
NFR-02	First Contentful Paint < 1.8 s on mid-range Android + 4G	Must
NFR-03	Lighthouse Performance score ≥ 90 (mobile)	Must
NFR-04	Bangla font correctly loaded and rendered	Must
NFR-05	Basic WCAG 2.1 AA accessibility (contrast, focus, labels)	Should
NFR-06	SEO-friendly (Next.js Metadata API, semantic HTML)	Must
NFR-07	Graceful loading, empty and error states everywhere	Must
NFR-08	Cart and auth state persist across page reloads	Must
NFR-09	Design system uses Blue theme + Inter + Serif fonts	Must
NFR-10	Codebase follows TypeScript strict mode and clean architecture	Must

5. UI/UX Requirements

5.1 Brand & Design System

- Project Name: mediShop
- Primary Theme Color: Blue (#1D4ED8 / Blue-700)
- Body Font: Inter (clean, modern, highly legible)
- Title / Header Font: Roboto Slab (Serif) for professional authority
- Accent color for discounts & important CTAs: Amber / Orange

5.2 Localization & Language

- Bangla-first experience. Default language is Bangla; English toggle available.
- All primary buttons, labels, error messages and empty states in clear Bangla.
- Medicine names preferably shown with both Bangla generic and English brand names.

5.3 Interaction & Accessibility

- Thumb-friendly touch targets (minimum 44×44 px).
- Clear visual hierarchy: Hero → Trust → Categories → Products.
- Skeleton loaders for every asynchronous section.
- Helpful empty states and friendly Bangla error messages.
- Micro-interactions on Add-to-Cart, button presses, drawer open/close.

6. External Interface Requirements

6.1 User Interfaces

All screens must be fully responsive. Primary design target is mobile (360–430 px width). Desktop enhancements (sidebar, multi-column grids) degrade gracefully.

6.2 Software Interfaces (Phase-1)

Phase-1 uses mock data modules. Future integration points: REST/GraphQL APIs for products, categories, auth, cart, orders, prescription upload, and payment initiation.

6.3 Communication Interfaces

Floating “Call to Order” and “WhatsApp Order” buttons deep-link to the business phone number and WhatsApp Business account respectively.

7. System Architecture & Technology Stack

7.1 High-Level Frontend Architecture

Next.js App Router is used. Server Components where beneficial; Client Components for interactive parts (cart, modals, filters). TanStack Query owns all server/async state. Redux Toolkit owns pure client state (cart contents, auth UI flags, modal open state).

7.2 Recommended Folder Structure

```
src/
├── app/                                # Next.js App Router
│   ├── (public)/
│   │   ├── page.tsx                  # Homepage
│   │   ├── products/page.tsx
│   │   ├── category/[slug]/page.tsx
│   │   ├── search/page.tsx
│   │   ├── cart/page.tsx
│   │   ├── checkout/page.tsx
│   │   └── product/[id]/page.tsx
│   ├── layout.tsx
│   └── globals.css
├── components/
│   ├── layout/                      # Navbar, Footer, CategorySidebar
│   ├── home/                       # HeroSlider, TrustBadges, ProductSection...
│   ├── auth/                       # AuthModal
│   ├── product/                   # ProductCard, FilterSidebar, SortDropdown
│   ├── cart/                      # CartDrawer, CartItem
│   └── ui/                        # shadcn/ui primitives
├── features/
│   ├── auth/
│   ├── cart/
│   └── products/
├── store/                          # Redux store + slices
├── hooks/                          # TanStack Query hooks
├── lib/
├── types/
└── mocks/                          # Realistic mock data for Phase-1
```

7.3 State Management Strategy

- Server / Async state (products, categories, search results) → TanStack Query
- Client state (cart items, auth modal visibility, UI flags) → Redux Toolkit
- Form state → React Hook Form + Zod validation

8. Data Requirements

Phase-1 relies on well-structured mock data that closely resembles the future API contracts. Key entities:

- Product (id, nameBn, nameEn, brand, price, mrp, discountPercent, image, categoryIds, requiresPrescription, stock)
- Category (id, nameBn, nameEn, icon, slug, parentId)
- CartItem (productId, quantity)
- User (id, name, phone, email) – mock only
- Address, Order (for checkout UI simulation)

9. Phase-wise Delivery Plan

9.1 Phase-1 – Frontend Design & Interactive Prototype (Current)

Deliverables that must be demo-ready for the client:

- Complete Homepage (Hero, Category Sidebar, Trust, Product sections, Prescription CTA)
- Sticky Navbar with Search, Cart icon + badge, Account/Login
- Authentication Modal (Login / Register UI)
- Cart Drawer + full Cart Page
- Checkout Page with address, delivery options, payment method selection and mock payment
- Product Listing pages with full Filter Sidebar + Sorting
- Product Card with working Add-to-Cart
- Floating Call & WhatsApp buttons
- Fully responsive (mobile + desktop)
- Skeleton loaders, empty states, basic error handling
- Bangla content support + Blue theme + Inter / Serif typography

9.2 Later Phases (High-level)

- Real backend integration & authentication
- Real payment gateway (bKash, Nagad, etc.)
- Prescription verification workflow & order tracking
- Doctor video consultation module
- Admin panel, inventory, analytics
- PWA / React Native mobile apps

10. Acceptance Criteria (Phase-1)

The Phase-1 delivery will be considered accepted when:

1. Client can open the application on both mobile and desktop browsers without broken layouts.
2. All major sections of the Homepage are visible, interactive and match the agreed visual direction (Blue theme, Inter + Serif).
3. Login / Register modal opens and closes correctly; Navbar reflects logged-in vs guest state (mock).
4. Products can be added to and removed from the cart; badge count updates; Cart Drawer and Cart Page work.
5. Checkout page can be completed end-to-end with mock payment success/failure paths.
6. Product listing pages support filtering by the defined facets and all sorting options.
7. Search returns results and respects the same filter/sort controls.
8. Design feels more modern, trustworthy and Bangladesh-optimized than the reference MedEasy site.
9. Code is written in TypeScript, uses the agreed stack, and follows clean component architecture.
10. No critical console errors; basic loading and empty states are present.
11. Project is branded as mediShop throughout the UI.

11. Future Phases (Out of Scope for Phase-1)

The following capabilities are explicitly out of scope for the current frontend Phase-1 but are expected in subsequent releases:

- Production backend APIs and database
- Real user registration, OTP verification and session management
- Actual payment gateway integration and reconciliation
- Pharmacist prescription verification dashboard and workflow
- Order management, status tracking and delivery partner integration
- Doctor listing, scheduling and video consultation
- Health records and prescription history
- Admin panel (products, inventory, orders, users, analytics)
- Push notifications, SMS/WhatsApp transactional messages
- Native mobile applications or advanced PWA features
- Advanced personalization and chronic-patient auto-refill

Appendix A: Quick Start Commands

Project Initialization

```
npx create-next-app@latest . --typescript --tailwind --eslint --app --src-dir --import-alias "@/*" --yes

npx shadcn@latest init

npm install @tanstack/react-query @reduxjs/toolkit react-redux \
  react-hook-form zod @hookform/resolvers lucide-react \
  framer-motion sonner
```

Design Tokens (Recommended)

```
Primary:      #1D4ED8  (Blue-700)
Primary Dark:#1E3A8A  (Blue-900)
Accent:       #F59E0B  (Amber-500)
Body Font:    Inter
Title Font:   Roboto Slab (or any clean Serif)
```

End of Software Requirements Specification — Version 1.1
mediShop • Blue Theme • Inter + Roboto Slab
This document is ready for development kickoff and client review.